

Supplementary Material 1: Supplementary Information

Overlapping roots: a transmission-tree framework for population-level contact-tracing dynamics

Journal: Mathematical Biosciences

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Supplementary Material 1

Model construction details, analytical and numerical details for the quasi-steady solution, the perturbative construction, technical results and the large-population ensemble protocol, the finite-population agent-based benchmark, closure validation, and reproducibility files

Associated supplementary files. Supplementary Material 2 (`supplementary_material_2.xlsx`) contains all 780 dimensional parameterizations, the published trajectory-error and convergence summaries, and deterministic-reference and event-term checks. Supplementary Material 3 (`supplementary_material_3.zip`) contains the available executable and audit code, numerical tables and data, event-term verification data, figure-generation files, and software requirements.

S1 Model construction details

These derivations underlie Section 2 of the main text.

S1.1 Proportional-allocation benchmark

The first-generation child-count classes determine the total forward-tracing loss but not the classes from which traced individuals are removed (main-text Section 2.2). A benchmark that closes the system by assumption supposes traced individuals are distributed across the child-count classes in the same proportions as the infectious population as a whole. This is an illustrative closure assumption and is *not* implied by the active-transmission-genealogy construction. Under it, the fraction of tracing-induced removals assigned to class r is I_r/I , so for $I > 0$

$$\left. \frac{dI_r}{dt} \right|_{\text{forward,prop}} = -\gamma_c p_f M_1 \frac{I_r}{I}. \quad (\text{S1})$$

Summing over r recovers the total forward-tracing loss $-\gamma_c p_f M_1$, since $\sum_{r \geq 0} I_r = I$. Multiplying by r and summing gives the corresponding contribution to M_1 ,

$$\left. \frac{dM_1}{dt} \right|_{\text{forward,prop}} = -\gamma_c p_f M_1 \sum_{r \geq 0} r \frac{I_r}{I} = -\gamma_c p_f \frac{M_1^2}{I}. \quad (\text{S2})$$

Combining with the transmission and primary-removal contributions gives a system closed in S , I , and M_1 alone:

$$\frac{dS}{dt} = -\beta SI, \quad (\text{S3})$$

$$\frac{dI}{dt} = \beta SI - \Gamma I - \gamma_c p_f M_1, \quad (\text{S4})$$

$$\frac{dM_1}{dt} = \beta SI - 2\Gamma M_1 - \gamma_c p_f \frac{M_1^2}{I}, \quad (\text{S5})$$

where the proportional term is defined for $I > 0$ and set to zero at $I = 0$, where necessarily $M_1 = 0$. The population-level loss $-\gamma_c p_f M_1$ follows from the tracing mechanism itself; the term $-\gamma_c p_f M_1^2/I$ does not, and is a consequence of the proportional-allocation assumption. The depth-resolved coordinates of main-text Section 2.3 avoid it.

S1.2 Complete rooted configurations and population classes

Let $\xi_i(t)$ denote the complete active-descendant configuration associated with root i , and $\mathcal{X}$ the set of possible rooted configurations. For $x \in \mathcal{X}$, define

$$I_x(t) = \sum_{i \in \mathcal{A}(t)} \mathbf{1}_{\{\xi_i(t)=x\}}, \quad (\text{S6})$$

the number of infectious roots whose rooted views have configuration x . The full collection $\{I_x(t)\}_{x \in \mathcal{X}}$ is a population-level description retaining the complete active genealogical information carried by the rooted views; the first-generation classes I_r are obtained by grouping configurations according to their numbers of active children. If $d_k(x)$ denotes the number of active depth- k descendants in configuration x , then

$$M_k(t) = \sum_{x \in \mathcal{X}} d_k(x) I_x(t), \quad (\text{S7})$$

which agrees with the definition by summation over rooted views. The depth coordinates M_k therefore arise as aggregates of the complete genealogy-aware classes I_x , rather than as independently imposed moment variables. Evolving the complete collection is neither analytically convenient nor necessary for the tracing quantities considered here.

S2 Analytical and numerical details for the QSS solution

This section provides the admissibility argument, numerical evaluation procedure, and stochastic comparison protocol supporting Section 3 of the main text. The notation and main-text equation numbers are retained below.

S2.1 Existence and admissibility of the implicit QSS solution

For $0 < c \leq 1$, define

$$F_{R,c}(u) = \sqrt{\frac{R}{c}} \frac{I_{R-cu+1}(2\sqrt{cR})}{I_{R-cu}(2\sqrt{cR})}, \quad 0 \leq u \leq 1. \quad (\text{S8})$$

The self-consistency condition in main-text Eq. (3.8) is $F_{R,c}(u) = u$. Since $R - cu > -1$ for $R > 0$, $0 < c \leq 1$, and $0 \leq u \leq 1$, and since $2\sqrt{cR} > 0$, the modified Bessel functions appearing in Eq. (S8) are positive and continuous. Hence $F_{R,c}$ is continuous and positive on $[0, 1]$.

For the positive decaying ratio sequence corresponding to a fixed trial value u , main-text Eq. (3.5) can be written as

$$r_k(u) = \frac{R}{R + k - cu + cr_{k+1}(u)}. \quad (\text{S9})$$

At $k = 1$, $F_{R,c}(u) = r_1(u)$. Therefore,

$$F_{R,c}(1) = \frac{R}{R + 1 - c + cr_2(1)} < 1, \quad (\text{S10})$$

where the inequality is strict because $r_2(1) > 0$. Since $F_{R,c}(0) > 0$, the continuous function $F_{R,c}(u) - u$ is positive at $u = 0$ and negative at $u = 1$. The intermediate value theorem therefore gives at least one self-consistent solution

$$m_1 \in (0, 1). \quad (\text{S11})$$

At any such solution, the denominator in Eq. (S9) exceeds R : indeed, $k - cm_1 > 0$ for every $k \geq 1$, and $cr_{k+1} > 0$. Hence

$$0 < r_k < 1, \quad k \geq 1. \quad (\text{S12})$$

Since $m_k = m_{k-1}r_k$ and $m_0 = 1$, it follows that

$$1 = m_0 > m_1 > m_2 > \cdots > 0. \quad (\text{S13})$$

For $k \geq 2$, $cm_1 < 1$ and $cr_{k+1} > 0$, so Eq. (S9) also gives

$$r_k < \frac{R}{R + k - 1}. \quad (\text{S14})$$

Consequently,

$$0 < m_k = m_1 \prod_{j=2}^k r_j < m_1 \prod_{j=2}^k \frac{R}{R + j - 1} \rightarrow 0 \quad (k \rightarrow \infty). \quad (\text{S15})$$

Thus every positive self-consistent solution of main-text Eq. (3.8) defines a genealogically admissible descendant sequence.

When $c = 0$, main-text Eq. (3.5) becomes explicit:

$$r_k(R, 0) = \frac{R}{R + k}, \quad m_k(R, 0) = \prod_{j=1}^k \frac{R}{R + j}, \quad (\text{S16})$$

and therefore $m_1(R, 0) = R/(R + 1)$. For each tested $c > 0$, the scalar root was obtained independently. Across the tested parameter grid, the selected roots vary continuously from the no-tracing value and thereby identify the positive branch used in main-text Section 3. This selection does not require a claim of global uniqueness for all mathematical roots of main-text Eq. (3.8).

S2.2 Numerical evaluation of the implicit solution

The primary calculation solves the scalar residual

$$G_{R,c}(u) = F_{R,c}(u) - u \quad (\text{S17})$$

for $u \in (0, 1)$. At $c = 0$, the explicit solution is $u = R/(R + 1)$. For each $c > 0$, the root was obtained independently by Brent's safeguarded method on $[0, 1]$, using absolute and relative tolerances of 10^{-14} . Across the tested grid, the resulting roots varied continuously from the no-tracing value and remained on the selected positive branch.

Modified-Bessel ratios were evaluated with exponentially scaled functions. Because the numerator and denominator in main-text Eq. (3.8) have the same argument, the common exponential factor cancels, avoiding overflow or underflow while preserving the required ratio (Amos 1974). For $c > 0$, once m_1 was obtained, the descendant coordinates were calculated from

$$m_k = \left(\frac{R}{c}\right)^{k/2} \frac{I_{R-cm_1+k}(2\sqrt{cR})}{I_{R-cm_1}(2\sqrt{cR})}, \quad k \geq 1. \quad (\text{S18})$$

At $c = 0$, the explicit expression in Eq. (S16) was used.

As an independent consistency calculation, the continued fraction was evaluated by backward recursion. For a terminal depth K , set

$$r_{K+1}^{(K)} = 0 \quad (\text{S19})$$

and compute

$$r_k^{(K)} = \frac{R}{R + k - cm_1 + cr_{k+1}^{(K)}}, \quad k = K, K-1, \dots, 1. \quad (\text{S20})$$

The truncated self-consistency equation $r_1^{(K)} = m_1$ was solved separately at $K = 40$ and $K = 80$. Over the main-text Fig. 2 grid, the maximum fixed-point residual of the scaled-Bessel calculation was below 3×10^{-15} ; the largest difference between the $K = 40$ and $K = 80$ values of m_1 , m_2 , and m_3 was below 10^{-15} , and the largest difference between the scaled-Bessel and $K = 80$ continued-fraction calculations was below 10^{-15} . The verification script and cellwise numerical results are supplied in Supplementary Material 3.

S2.3 Constant-pool stochastic comparison protocol

The constant-pool simulations use

$$R \in \{1, 1.5, 2, 4\}, \quad c \in \{0, 0.25, 0.5, 0.75, 1\}, \quad (\text{S21})$$

with dimensionless rates

$$\Gamma = 1, \quad \gamma = 0, \quad \gamma_c = 1, \quad p_f = c. \quad (\text{S22})$$

The susceptible pool is held constant, so transmission occurs at total rate RI . Tracing is active from $\tau = 0$. Each realization begins with 160 unrelated infectious roots, and 120 independent realizations are generated for every parameter pair over $0 \leq \tau \leq 3.25$.

For each active realization, the normalized descendant coordinate is

$$m_k(\tau) = \frac{M_k(\tau)}{I(\tau)}, \quad I(\tau) > 0. \quad (\text{S23})$$

A coordinate is left undefined after the realization no longer has an active infectious population; no stopped or terminal genealogy is carried forward.

The normalized coordinates were evaluated on the uniform output grid used to generate main-text Fig. 2. Because the output spacing is uniform, equal weighting of the retained observations gives the corresponding discrete time average. For replicate ℓ and depth k , let $\mathcal{T}_{k,\ell}$ be the recorded times in the prespecified interval

$$3 \leq \tau \leq 3.25 \quad (\text{S24})$$

for which $I(\tau) > 0$ and $m_k(\tau)$ is defined. The replicate-level summary is

$$m_{k,\ell}^{\text{ABM}} = \frac{1}{|\mathcal{T}_{k,\ell}|} \sum_{\tau_j \in \mathcal{T}_{k,\ell}} m_k(\tau_j), \quad (\text{S25})$$

provided $|\mathcal{T}_{k,\ell}| > 0$. Thus, values after extinction are never imputed or carried forward; a replicate is excluded for depth k only when it has no defined observation in the averaging interval.

Let $\mathcal{V}_k$ denote the replicates with a defined summary and $n_k = |\mathcal{V}_k|$. The signed difference plotted in Fig. 2a–c is

$$e_{k,\text{QSS}}^{\text{ABM}} = \frac{1}{n_k} \sum_{\ell \in \mathcal{V}_k} m_{k,\ell}^{\text{ABM}} - m_k^{\text{QSS}}, \quad k = 1, 2, 3. \quad (\text{S26})$$

Percentile 95% confidence intervals were obtained from 2,000 bootstrap resamples of complete replicate-level summaries; recorded time points within a realization were therefore never resampled as independent observations. For the trajectory panels, complete replicate trajectories were resampled and the ensemble mean was recalculated at each recorded time to obtain pointwise 95% confidence intervals. The representative panels use $R = 2$ and $c = 1$; the pale-yellow region marks the averaging interval and the horizontal dashed line gives the selected QSS target.

S3 Perturbative construction of the QSS approximation

This section derives the explicit low-order approximation to m_1 quoted in Section 3.2 of the main text, and reports its accuracy against the selected QSS branch.

S3.1 Rescaled QSS recurrence

At $c = 0$, the QSS recurrence has the no-tracing solution

$$m_{k,0} = \prod_{j=1}^k \frac{R}{R+j}, \quad k \geq 1. \quad (\text{S27})$$

Define

$$\hat{m}_k = \frac{m_k}{m_{k,0}}, \quad b_k = \frac{R+1}{R+k}, \quad \chi_k = \frac{R+1}{R+k+1}, \quad \varepsilon = \frac{cR}{(R+1)^2}. \quad (\text{S28})$$

Substitution into the QSS recurrence gives

$$\hat{m}_k = \hat{m}_{k-1} + \varepsilon b_k (\hat{m}_1 \hat{m}_k - \chi_k \hat{m}_{k+1}), \quad k \geq 1. \quad (\text{S29})$$

For $R > 0$ and $0 \leq c \leq 1$,

$$0 \leq \varepsilon \leq \frac{R}{(R+1)^2} \leq \frac{1}{4}. \quad (\text{S30})$$

The final inequality follows from $(R-1)^2 \geq 0$ and is sharp at $R = 1$, $c = 1$.

S3.2 Coefficient recursion and finite genealogical dependence

Use the regular expansion

$$\hat{m}_k = 1 + \sum_{n \geq 1} a_{k,n} \varepsilon^n, \quad a_{k,0} = 1, \quad a_{0,n} = 0 \quad (n \geq 1). \quad (\text{S31})$$

Matching equal powers of ε in Eq. (S29) yields

$$a_{k,n} = a_{k-1,n} + b_k \left[\sum_{j=0}^{n-1} a_{1,j} a_{k,n-1-j} - \chi_k a_{k+1,n-1} \right], \quad n \geq 1. \quad (\text{S32})$$

The recursion couples adjacent descendant depths while lowering the perturbative order by one in the deeper term. Consequently,

$$a_{1,p} \longrightarrow a_{2,p-1} \longrightarrow \cdots \longrightarrow a_{p,1} \longrightarrow a_{p+1,0}.$$

The deepest required quantity is the known no-tracing coefficient $a_{p+1,0} = 1$. Hence every fixed perturbative order is determined by a finite genealogical calculation.

For $k = 1$,

$$a_{1,1} = \frac{1}{R+2}, \quad (\text{S33})$$

$$a_{1,2} = \frac{7+2R-R^2}{(R+2)^2(R+3)}. \quad (\text{S34})$$

Therefore,

$$m_1^{[p]}(R, c) = \frac{R}{R+1} \left[1 + \sum_{n=1}^p a_{1,n}(R) \left(\frac{cR}{(R+1)^2} \right)^n \right]. \quad (\text{S35})$$

The formulas used in Fig. S1 are

$$m_1^{[0]} = \frac{R}{R+1}, \quad (\text{S36})$$

$$m_1^{[1]} = \frac{R}{R+1} \left[1 + \frac{cR}{(R+1)^2(R+2)} \right], \quad (\text{S37})$$

$$m_1^{[2]} = \frac{R}{R+1} \left[1 + \frac{cR}{(R+1)^2(R+2)} + \frac{7+2R-R^2}{(R+2)^2(R+3)} \left(\frac{cR}{(R+1)^2} \right)^2 \right]. \quad (\text{S38})$$

S3.3 Local finite-depth basis

Consider a finite truncation retaining $\hat{m}_1, \dots, \hat{m}_K$ and an analytic terminal prescription $\hat{m}_{K+1} = T_K(\varepsilon)$ satisfying $T_K(0) = 1$. The truncated QSS equations are analytic in the retained moments and ε . At $\varepsilon = 0$, Eq. (S29) reduces to $\hat{m}_k = \hat{m}_{k-1}$, so the no-tracing solution is $\hat{m}_k = 1$. The Jacobian with respect to the retained moments is lower bidiagonal with unit diagonal and is therefore nonsingular. Standard analytic implicit-function and regular-perturbation arguments consequently give a locally unique analytic branch whose coefficients satisfy Eq. (S32) (Krantz and Parks 2002; Murdock 1999).

This is a local statement for each fixed finite truncation. The main-text claim does not require a global convergence theorem for the complete infinite series; instead, the accuracy of the explicit low-order formulas is assessed directly against the selected QSS branch.

S3.4 Numerical evaluation for the perturbative accuracy figure

The selected QSS branch was evaluated by solving the scalar fixed-point equation generated by a backward continued fraction. Fig. S1 was generated using continued-fraction depth 80 and a bracketed scalar solve on $0 \leq m_1 \leq 1$. The error curves were evaluated at $R \in \{1, 1.5, 2, 4\}$ and at 1001 equally spaced values of c on $[0, 1]$. For retained order p ,

$$e_p^{\text{pert}}(R, c) = \left| m_1^{[p]}(R, c) - m_1^{\text{QSS}}(R, c) \right|. \quad (\text{S39})$$

The maximum errors over the c -grid are summarized in Table S1.

Table S1 Maximum absolute perturbative errors e_p^{pert} over $c \in [0, 1]$ for the parameter values used in Fig. S1.

R	$p = 0$	$p = 1$	$p = 2$
1.0	5.0374326×10^{-2}	8.7076597×10^{-3}	1.7632153×10^{-3}
1.5	4.6553438×10^{-2}	5.4105804×10^{-3}	5.5180492×10^{-4}
2.0	3.9907541×10^{-2}	2.8705039×10^{-3}	1.0154532×10^{-5}
4.0	2.1111821×10^{-2}	2.2151187×10^{-4}	1.4024202×10^{-4}

For every displayed R and p , the maximum occurred at $c = 1$. The signed $p = 2$ error for $R = 2$ crossed zero at $c = 0.8942672386$.

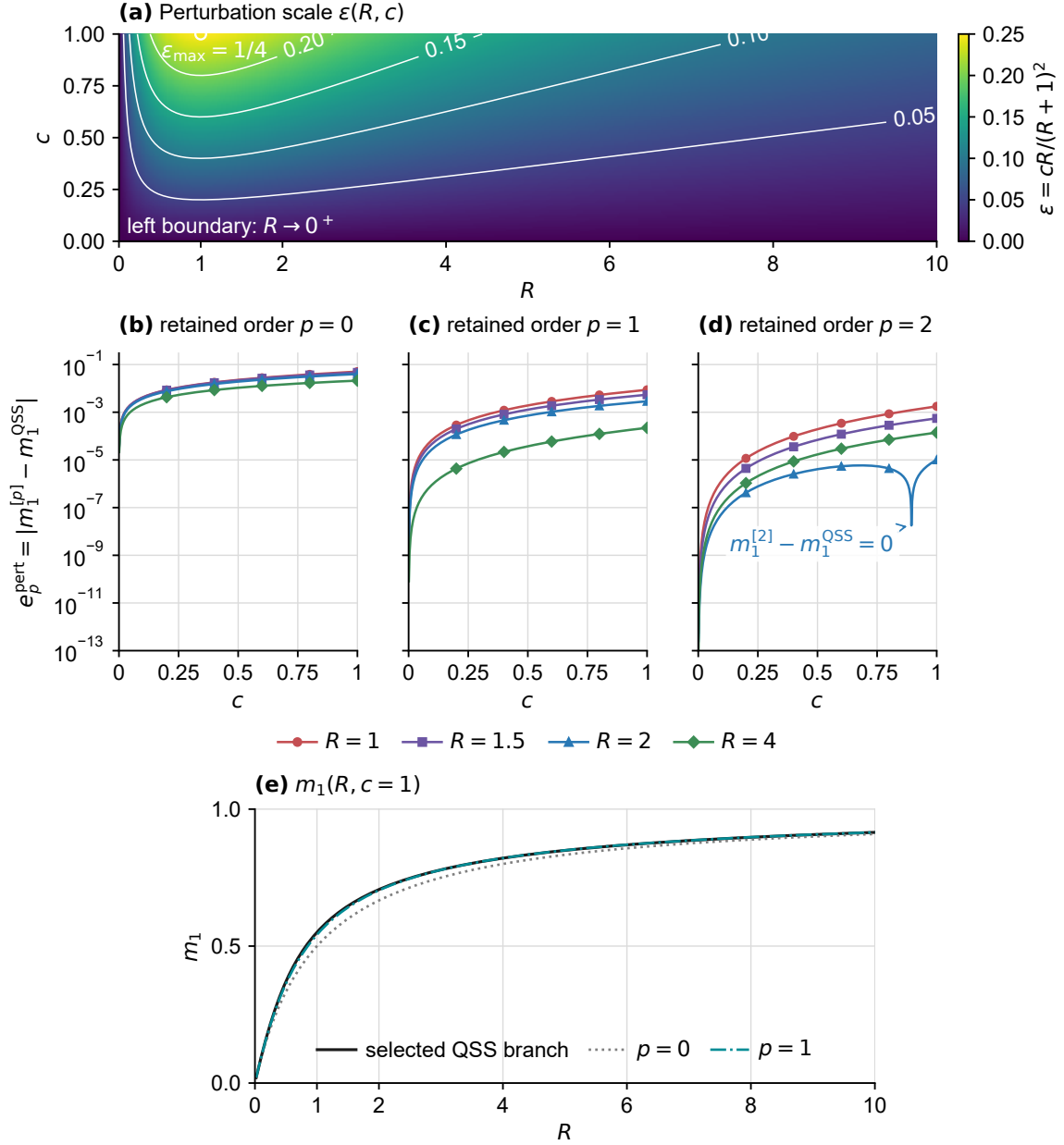


Fig. S1 Effective perturbation scale and numerical accuracy of the explicit QSS approximation. (a) The perturbation scale $\varepsilon = cR/(R+1)^2$ over $0 < R \leq 10$ and $0 \leq c \leq 1$. It satisfies $0 \leq \varepsilon \leq 1/4$, with equality at $R = 1$ and $c = 1$; the displayed left boundary represents the limit $R \rightarrow 0^+$. (b–d) Absolute errors $e_p^{\text{pert}} = |m_1^{[p]} - m_1^{\text{QSS}}|$ for retained orders $p = 0, 1, 2$, respectively, at $R = 1, 1.5, 2$, and 4 . Errors vanish at $c = 0$ and are omitted from the logarithmic axes. The narrow minimum in the $R = 2$ curve in panel (d) occurs where the signed second-order error crosses zero near $c = 0.894$. (e) The selected QSS branch and the $p = 0$ and $p = 1$ approximations along $c = 1$. The first-order approximation is already nearly coincident with the selected branch over the displayed range. These errors compare the explicit perturbative formulas against the selected QSS branch itself; they are not errors of the QSS branch relative to the underlying epidemic process, which is assessed separately in main-text Section 3.3.

The QSS solver was checked over $R \in [0.05, 10]$ and $c \in [0.01, 1]$ on a 124×100 diagnostic grid. Depths 40, 80, and 160 agreed to double-precision resolution. At depth 160, the maximum fixed-point residual was 2.33×10^{-15} , and the maximum discrepancy between the continued-fraction evaluation and the evaluation based on the modified-Bessel representation was 4.44×10^{-15} .

S4 Technical results and large-population ABM ensemble protocol

These technical results and the ensemble protocol support the epidemiological observables and the control boundary of Section 4 of the main text.

S4.1 Equality of the growth and replacement thresholds

On the selected constant-pool QSS branch, the normalized moments satisfy

$$0 = R(m_{k-1} - m_k) - km_k + c(m_1 m_k - m_{k+1}), \quad k \geq 1, \quad (\text{S40})$$

with $m_0 = 1$. Define

$$B(R, c) = 1 + \sum_{k \geq 1} \frac{(-c)^k}{(k+1)!} m_k(R, c). \quad (\text{S41})$$

The factorial weights make all sums below absolutely convergent. Multiply Eq. (S40) by $(-c)^k/(k+1)!$ and sum over $k \geq 1$. After index shifts, the transmission terms generate the series in $\mathcal{R}_{\text{time}}$, while the remaining terms collect into gB , giving

$$\mathcal{R}_{\text{time}}(R, c) - 1 = g(R, c)B(R, c). \quad (\text{S42})$$

For completeness, the weighted sum can be written directly as

$$0 = \sum_{k \geq 1} \frac{(-c)^k}{(k+1)!} [R(m_{k-1} - m_k) - km_k + c(m_1 m_k - m_{k+1})] \quad (\text{S43})$$

$$= \mathcal{R}_{\text{time}} - 1 - gB. \quad (\text{S44})$$

Moreover, Eq. (S41) is an alternating series with strictly decreasing term magnitudes because

$$\frac{c^{k+1} m_{k+1}/(k+2)!}{c^k m_k/(k+1)!} = \frac{c}{k+2} \frac{m_{k+1}}{m_k} \leq \frac{1}{k+2} < 1. \quad (\text{S45})$$

Hence

$$B(R, c) \geq 1 - \frac{cm_1}{2} > \frac{1}{2}. \quad (\text{S46})$$

Therefore

$$g(R, c) = 0 \iff \mathcal{R}_{\text{time}}(R, c) = 1, \quad (\text{S47})$$

although the two quantities have different numerical values and interpretations away from the boundary.

S4.2 Finite-depth evaluation of $\mathcal{R}_{\text{time}}$

Retaining genealogical coordinates through m_{K-1} , define

$$\mathcal{R}_{\text{time}}^{(K)}(R, c) = R \left[1 + \sum_{k=0}^{K-1} \frac{(-c)^{k+1}}{(k+2)!} m_k(R, c) \right]. \quad (\text{S48})$$

Because m_k is nonincreasing and $0 \leq c \leq 1$, the omitted tail is alternating with decreasing magnitude. Its error is bounded by the first omitted term:

$$\left| \mathcal{R}_{\text{time}} - \mathcal{R}_{\text{time}}^{(K)} \right| \leq R \frac{c^{K+1}}{(K+2)!} m_K \leq R m_K. \quad (\text{S49})$$

The factorial factor makes the lifetime replacement quantity substantially less sensitive to deep-tail error than an unweighted moment sum.

S4.3 Propagation of QSS approximation errors

If an approximation $m_1^{[p]}$ is used in the growth rate, then

$$g^{[p]} - g^{\text{QSS}} = -c \left(m_1^{[p]} - m_1^{\text{QSS}} \right). \quad (\text{S50})$$

If approximate moments $m_k^{[p]}$ are used through depth $K - 1$, then

$$\left| \mathcal{R}_{\text{time}}^{[p,K]} - \mathcal{R}_{\text{time}}^{\text{QSS}} \right| \leq R \sum_{k=1}^{K-1} \frac{c^{k+1}}{(k+2)!} \left| m_k^{[p]} - m_k^{\text{QSS}} \right| \quad (\text{S51})$$

$$+ R \frac{c^{K+1}}{(K+2)!} m_K^{\text{QSS}}. \quad (\text{S52})$$

The first term is the error in the retained coordinates and the second is the unresolved genealogical tail. This analysis concerns approximation of the selected QSS branch; it is distinct from finite-population ABM–QSS discrepancy.

S4.4 Large-population constant-pool ABM ensemble protocol

The ABM uses the same transmission, primary-removal, detection, and instantaneous one-step forward-tracing events as the deterministic hierarchy. The susceptible pool is held constant in the transmission rate. The decomposition used for main-text Fig. 3 is

$$\frac{\gamma}{\Gamma} = 0, \quad \frac{\gamma_c}{\Gamma} = 1, \quad p_f = c. \quad (\text{S53})$$

Thus, the deterministic prediction depends on R and c , while the ABM realizes the corresponding finite-population event system.

Table S2 Simulation design used for main-text Fig. 3.

Component	Cells	Design
Panels (a)–(b)	$R = 1, 1.5, 2, 4$; $c = 0, 0.25, 0.5, 0.75, 1$	300 independent paths per cell; prespecified burn-in and windows; whole-replicate bootstrap intervals.
Cohort follow-up	Same 20 cells	Complete follow-up of every entrant; no lifetime or required production interval is censored.
Panel (c)	$c = 0.25, 0.5, 0.75, 1$; two prespecified R -bracketing cells	Pooled growth estimates, linear interpolation, and point-wise whole-replicate bootstrap intervals.

For replicate r , let

$$P_r = \int_{\tau_{0,r}}^{\tau_{1,r}} I_r(\tau) d\tau \quad (\text{S54})$$

be the parent infectious person-time in the prespecified measurement or cohort-entry window. If $N_{+,r}$ is the number of new infections and $N_{-,r}$ is the number of infectious individuals removed by primary removal or tracing during the growth-measurement window, the pooled event/person-time estimator is

$$\hat{g} = \frac{\sum_r (N_{+,r} - N_{-,r})}{\sum_r P_r}. \quad (\text{S55})$$

Here $N_{-,r}$ counts removed individuals rather than removal-event occurrences, so tracing events that remove more than one infectious individual contribute their full population change.

Let $\mathcal{C}_r$ be the set of infections entering during the cohort window, and let L_{jr} be the complete remaining infectious lifetime of cohort member j . The pooled infectious-time replacement estimator

is

$$\hat{\mathcal{R}}_{\text{time}} = \frac{\sum_r \sum_{j \in \mathcal{C}_r} L_{jr}}{\sum_r P_r}. \quad (\text{S56})$$

The event system is continued along each path until every enrolled cohort member has been removed. The entry window, burn-in rule, and production sample size are fixed before the confirmatory runs.

Confidence intervals are obtained by resampling complete replicates and recomputing the pooled ratio in Eqs. (S55) and (S56). Serial observations within a replicate are not treated as independent. For panel (c), let $(R^-, \hat{g}^-)$ and $(R^+, \hat{g}^+)$ denote the two prespecified cells bracketing zero. The interpolated estimate is

$$\hat{R}_*(c) = R^- + \frac{-\hat{g}^-}{\hat{g}^+ - \hat{g}^-} (R^+ - R^-), \quad (\text{S57})$$

and each bootstrap resample recomputes the two pooled growth estimates and the interpolation.

S4.5 Analytical and numerical checks

The analytical curves use the same selected-QSS solver as Fig. S1 and main-text Fig. 3. The final audit gave a maximum fixed-point residual of 1.887×10^{-13} , a maximum accepted depth-doubling difference of 1.943×10^{-13} , and a maximum modified-Bessel/continued-fraction difference of 1.894×10^{-13} . The control-boundary values used in Fig. 4c are

c	0	0.25	0.50	0.75	1.00	(S58)
$R_*(c)$	1.00000000	1.13568648	1.29328143	1.47215210	1.66978036	

The threshold identity was checked on 8,888 parameter points. Across the 20 cohort cells, the maximum relative Monte Carlo standard error was 0.50%, the maximum analytical discrepancy was 0.67%, and 19 of 20 pointwise intervals contained their targets; all panel-(c) intervals contained the analytical critical curve. These comparisons assess large-population ensemble behavior of the same event system, not an unrelated stochastic model.

S5 Finite-population agent-based benchmark

This section documents the individual-level benchmark introduced in Section 2.4 of the main text and used for the comparisons reported in Sections 3.3, 4.2 and 5.2.

S5.1 The agent-based model

The finite-population agent-based model (ABM) simulates transmission, spontaneous removal, detection, and instantaneous one-step forward tracing on the active transmission forest. Individual identities and unique infector–infectee links are retained throughout each realization. Let $U^{\text{ABM}}(t)$ and $I^{\text{ABM}}(t)$ denote the susceptible and infectious counts in physical time. Transmission, spontaneous removal, and detection occur at total rates

$$\beta U^{\text{ABM}}(t) I^{\text{ABM}}(t), \quad \gamma I^{\text{ABM}}(t), \quad \gamma_c I^{\text{ABM}}(t),$$

respectively. At transmission, an infectious individual is selected uniformly as the infector, one susceptible individual becomes infectious, and the new case is recorded as an active child of its infector. At spontaneous removal, the selected individual leaves the active infectious set. At detection, the selected individual is removed and each active direct infectee is independently removed

with probability p_f ; traced removals do not initiate further tracing. Descendant-depth observables $M_k^{\text{ABM}}(t)$ can be computed directly from the active transmission forest at each observation time; the numerical trajectory comparison uses M_1^{ABM} . The process is simulated with an exact event-driven Gillespie algorithm.

The dimensional parameters determine

$$\Gamma = \gamma + \gamma_c, \quad \tau = \Gamma t, \quad R = \frac{\beta U_0}{\Gamma}, \quad c = \frac{\gamma_c p_f}{\Gamma}.$$

The process is simulated in physical time and reported on $0 \leq \tau \leq 5$. Detection operates throughout the comparison experiments, whereas forward tracing is activated at $\tau_{\text{on}} = 0.5$. The same activation schedule is imposed in the deterministic reference.

S5.2 Crossed dimensional parameterizations

The nondimensional parameters R and c do not uniquely determine the dimensional rates. We therefore cross

$$\Gamma \in \left\{ \frac{1}{4}, \frac{1}{2}, 1, 2, 4 \right\}$$

with three explicit detection–tracing decompositions. For prescribed R , c , U_0 , and Γ , the parameterizations are

Table S3 Crossed dimensional parameterizations for $0 \leq c < 1$

Decomposition	β	γ	γ_c	p_f
Less-frequent detection, complete tracing	$R\Gamma/U_0$	$\Gamma(1 - c)$	Γc	1
Intermediate	$R\Gamma/U_0$	$\Gamma(1 - c)/2$	$\Gamma(1 + c)/2$	$2c/(1 + c)$
Frequent detection, partial tracing	$R\Gamma/U_0$	0	Γ	c

Every row satisfies

$$\gamma + \gamma_c = \Gamma, \quad \frac{\beta U_0}{\Gamma} = R, \quad \frac{\gamma_c p_f}{\Gamma} = c.$$

At $c = 1$, the three rows coincide, and one parameterization is retained for each value of Γ . At $c = 0$, the three rows differ in the labeling of primary removal, but no tracing removal occurs.

The analysis uses $R \in \{2, 4, 6\}$ and $c \in \{0, 0.25, 0.5, 0.75, 1\}$. For each population scale, this gives 195 raw-rate protocols: 65 for each value of R . All raw-rate protocols are simulated directly. For $U_0 \in \{500, 1000, 2000\}$, one independent pool of 120 trajectories is generated per protocol for population-size convergence. At $U_0 = 8000$, 30 independent pools of 120 trajectories are generated for every protocol; the first prespecified pool (pool 0) is also used as the $U_0 = 8000$ point in the population-size comparison. The 30 pools give

$$195 \times 30 \times 120 = 702,000$$

ABM trajectories at the selected population scale.

Table S4 Global numerical settings

Setting	Value
Nondimensional grid	$R \in \{2, 4, 6\}$, $c \in \{0, 0.25, 0.5, 0.75, 1\}$
Dimensional rate scales	$\Gamma \in \{1/4, 1/2, 1, 2, 4\}$
Population scales	$U_0 \in \{500, 1000, 2000, 8000\}$, $I_0 = 0.02U_0$
Raw-rate protocols	195 per population scale; 780 parameterizations in total
Population convergence	One 120-realization pool per protocol; at $U_0 = 8000$, pool 0 of the 30 replicate-convergence pools is used
Replicate convergence at $U_0 = 8000$	30 independent 120-realization pools per raw-rate protocol
Observation grid	151 equally spaced points on $0 \leq \tau \leq 5$
Tracing activation	$\tau_{\text{on}} = 0.5$
High-depth reference	$K_{\text{ref}} = 40$, checked against $K = 80$
Deterministic solver	BDF; relative tolerance 10^{-10} , absolute tolerance 10^{-12} , maximum step 0.025
Pointwise confidence intervals	Normal approximation; conditional on $I(\tau) > 0$ for m_1
Master seed	20260804

S5.3 High-depth deterministic reference

For each (R, c) , the ABM ensemble mean is compared with a high-depth numerical solution retaining U , I , and $M_1, \dots, M_{K_{\text{ref}}}$, with $K_{\text{ref}} = 40$. The initial state is

$$U(0) = U_0, \quad I(0) = I_0, \quad M_k(0) = 0 \quad (k \geq 1).$$

At the terminal retained level, $M_{K_{\text{ref}}+1} = 0$ is imposed. High-depth trajectories are computed with the BDF method using relative tolerance 10^{-10} , absolute tolerance 10^{-12} , and maximum dimensionless step size 0.025. The intervals before and after $\tau_{\text{on}} = 0.5$ are integrated separately so that the change in the tracing terms is imposed exactly at the activation time. The structural bound $0 \leq M_{K_{\text{ref}}+1} \leq M_{K_{\text{ref}}}$ follows from Lemma 1 of the main manuscript, but adequacy is assessed independently by repeating the calculation with $K = 80$. Across the full grid and observation interval, the maximum absolute difference in $(U/U_0, I/U_0, M_1/U_0)$ was 2.99×10^{-10} .

S5.4 Ensemble-mean trajectory error and convergence

For realization ℓ , define

$$\mathbf{z}^{\text{ABM},(\ell)}(\tau) = \left(\frac{U^{\text{ABM},(\ell)}}{U_0}, \frac{I^{\text{ABM},(\ell)}}{U_0}, \frac{M_1^{\text{ABM},(\ell)}}{U_0} \right).$$

For a pool of n independent realizations, $\bar{\mathbf{z}}_n^{\text{ABM}} = n^{-1} \sum_{\ell=1}^n \mathbf{z}^{\text{ABM},(\ell)}$. With $\mathbf{z}^{\text{HD}}$ denoting the high-depth trajectory,

$$E_{\text{mean}} = \left[\frac{1}{T} \int_0^T \left\| \bar{\mathbf{z}}_n^{\text{ABM}}(\tau) - \mathbf{z}^{\text{HD}}(\tau) \right\|_2^2 d\tau \right]^{1/2}.$$

The integral is evaluated over $0 \leq \tau \leq 5$ at 151 equally spaced observation times using the trapezoidal rule. Extinct realizations remain at $I^{\text{ABM}} = M_1^{\text{ABM}} = 0$ and contribute to unconditional scaled-count means. Pointwise 95% confidence intervals in Figs. S3 and S4 are calculated as $\bar{x}(\tau) \pm 1.96s_x(\tau)/\sqrt{n_\tau}$. For U , I , and M_1 , $n_\tau = 120$. For $m_1 = M_1/I$, the mean, standard deviation, and confidence interval are calculated conditionally over realizations satisfying $I(\tau) > 0$, and n_τ is the number of contributing nonextinct realizations.

For population-size convergence, one 120-realization pool is used for each raw-rate protocol. For $U_0 \in \{500, 1000, 2000\}$, this pool is generated independently for the population-size calculation. At $U_0 = 8000$, the first prespecified pool (pool 0) among the 30 pools generated for replicate convergence is reused in the population-size comparison. For replicate convergence at $U_0 = 8000$, every raw-rate protocol has 30 independent master pools. Within each pool, the 120 trajectories are randomly permuted once, and the first n trajectories are used for $n \in \{1, 2, 5, 10, 20, 40, 80, 120\}$. Thus, each value of n , including $n = 120$, has

$$195 \times 30 = 5850$$

independent ensemble-error estimates. Estimates at different values of n are paired within a pool, but the 5850 estimates at any fixed n arise from independent pools. The representative trajectory plots in Figs. S3–S5 use the first prespecified independently generated pool (pool 0) for each displayed protocol; no pool was selected according to its agreement with the deterministic reference.

Table S5 Population-size convergence. Each row summarizes one 120-realization ensemble for each of the 195 raw-rate protocols. For $U_0 = 8000$, the first prespecified pool (pool 0) among the 30 independently generated pools is used.

U_0	I_0	Protocols	Estimates	Median	10th percentile	90th percentile
500	10	195	195	0.012926	0.006713	0.026509
1000	20	195	195	0.006759	0.002899	0.013034
2000	40	195	195	0.003335	0.001572	0.006732
8000	160	195	195	0.001337	0.000664	0.002717

Table S6 Replicate convergence at $U_0 = 8000$. Thirty independent 120-realization pools were generated for each of the 195 raw-rate protocols. Consequently, every value of n , including $n = 120$, is summarized by 5850 independent ensemble-error estimates.

n	Protocols	Pools/protocol	Estimates	Median	10th percentile	90th percentile
1	195	30	5850	0.012499	0.006548	0.027396
2	195	30	5850	0.008872	0.004631	0.019258
5	195	30	5850	0.005733	0.002934	0.012062
10	195	30	5850	0.003970	0.002051	0.008892
20	195	30	5850	0.002852	0.001492	0.006218
40	195	30	5850	0.002062	0.001051	0.004493
80	195	30	5850	0.001532	0.000775	0.003413
120	195	30	5850	0.001308	0.000654	0.002885

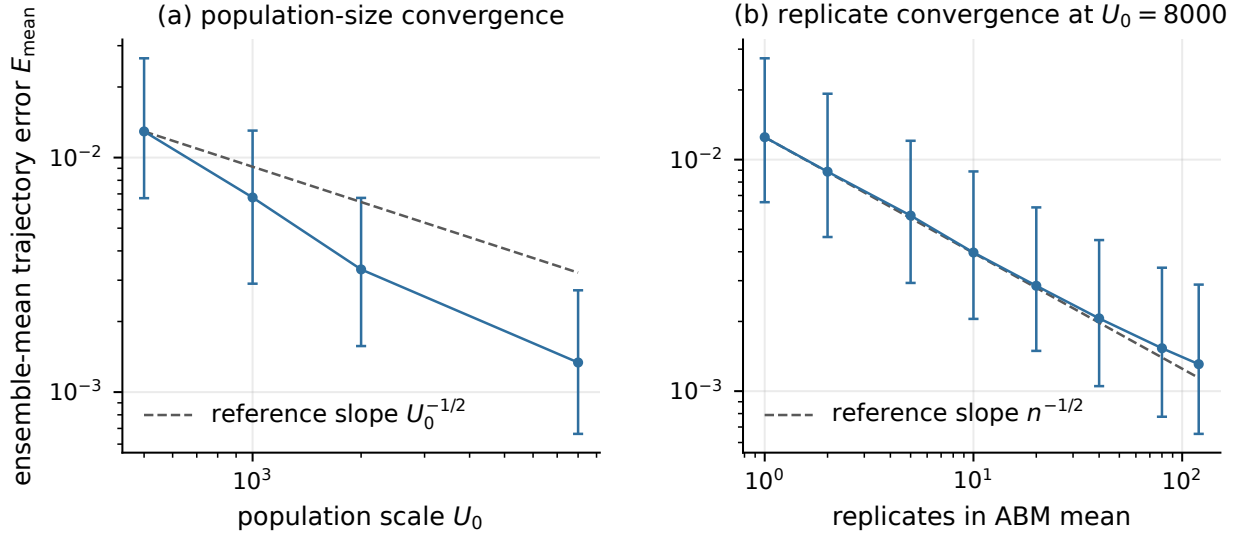
S5.5 Supplementary numerical results

Population-size convergence was systematic: increasing U_0 from 500 to 8000 reduced the median ensemble-mean trajectory error from 0.012926 to 0.001337, with a descriptive log–log slope of -0.82 over the tested population scales. On this basis, $U_0 = 8000$ was selected as the finite-population benchmark. It was the largest population examined, yielded discrepancies of order 10^{-3} , and remained computationally feasible for the full crossed parameter design. This is an empirically supported working scale rather than a universal threshold, and the reported discrepancies are empirical summaries rather than rigorous error bounds.

At $U_0 = 8000$, $I_0 = 160$, and $n = 120$, the 5850 independently generated ensembles had median trajectory error 0.001308, mean 0.001578, 90th percentile 0.002885, and maximum 0.006918. The descriptive log–log slope with replicate count was -0.48, close to the $n^{-1/2}$ reference scaling.

Table S7 Trajectory-error summary by nondimensional grid cell at $U_0 = 8000$ and $n = 120$. Each raw-rate protocol contributes 30 independently generated ensembles.

R	c	Protocols	Estimates	Median	Mean	90th percentile	Maximum
2	0	15	450	0.001786	0.002104	0.003983	0.006454
2	0.25	15	450	0.002028	0.002302	0.004118	0.006816
2	0.5	15	450	0.001761	0.002100	0.003910	0.006918
2	0.75	15	450	0.001658	0.001950	0.003545	0.006671
2	1	5	150	0.001613	0.001772	0.003014	0.005445
4	0	15	450	0.001225	0.001402	0.002466	0.004225
4	0.25	15	450	0.001294	0.001479	0.002607	0.004766
4	0.5	15	450	0.001303	0.001502	0.002474	0.004674
4	0.75	15	450	0.001368	0.001538	0.002493	0.006012
4	1	5	150	0.001545	0.001730	0.002964	0.004726
6	0	15	450	0.000992	0.001136	0.001870	0.003310
6	0.25	15	450	0.001012	0.001132	0.001950	0.003527
6	0.5	15	450	0.000995	0.001146	0.001988	0.003897
6	0.75	15	450	0.001006	0.001174	0.001999	0.003481
6	1	5	150	0.001046	0.001148	0.001792	0.003065

**Fig. S2** Convergence of the ABM ensemble-mean trajectory error. Panel (a) summarizes one 120-realization ensemble for each of 195 raw-rate protocols at each population scale; for $U_0 = 8000$, this is the first prespecified pool (pool 0) among the 30 independently generated pools. Panel (b) uses 30 independent pools per protocol at $U_0 = 8000$, yielding 5850 estimates for every n , including $n = 120$. Points show medians and error bars the 10th and 90th percentiles. Dashed lines indicate $U_0^{-1/2}$ and $n^{-1/2}$ reference scalings.**Table S8** Representative frequent-detection, partial-tracing errors shown in Fig. S3

c	E_U	E_I	E_{M_1}	E_{mean}
0	0.000722	0.000775	0.000586	0.001211
0.25	0.003258	0.001966	0.001638	0.004143
0.5	0.000898	0.000837	0.000900	0.001522
0.75	0.000602	0.000350	0.000313	0.000763
1	0.002752	0.001310	0.001063	0.003228

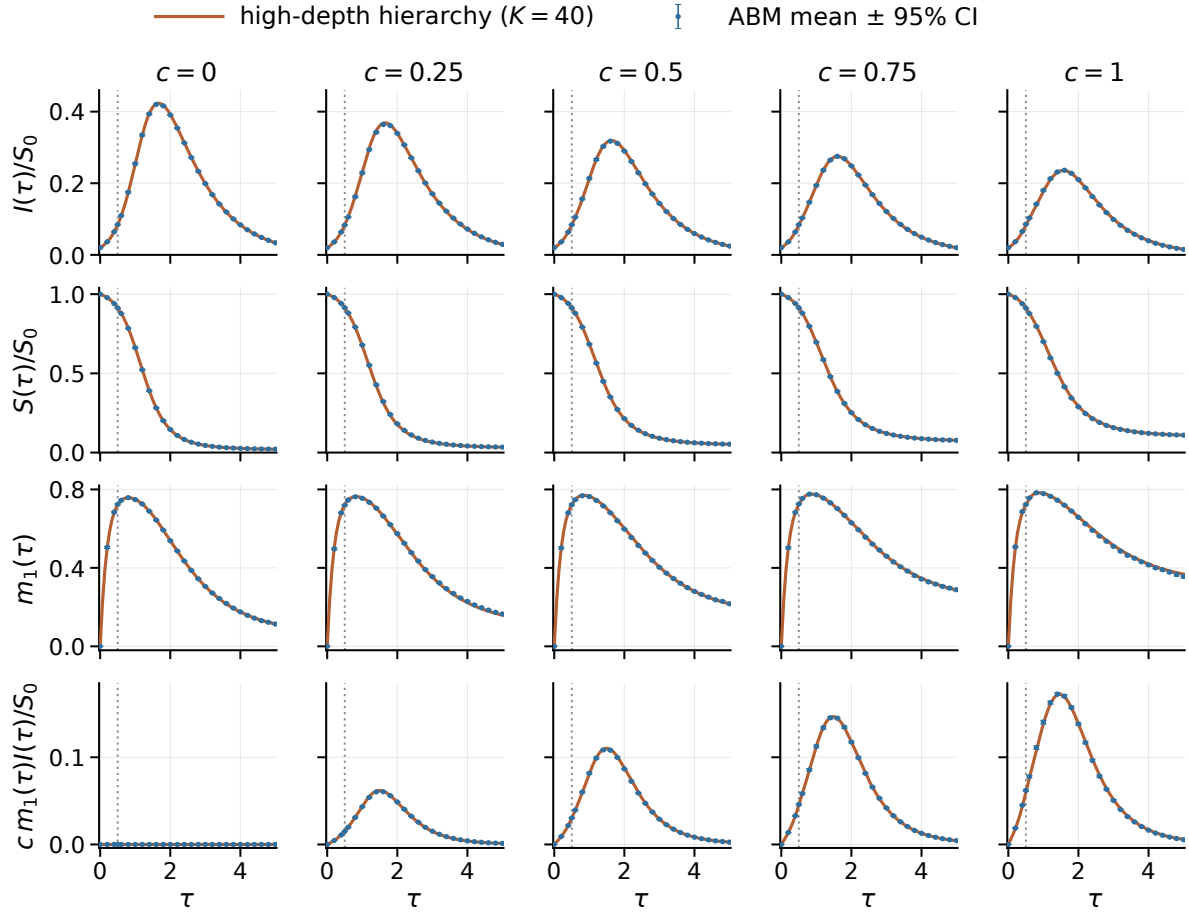


Fig. S3 Finite-population ABM–hierarchy comparison at $R = 4$, $U_0 = 8000$, $I_0 = 160$, and 120 realizations. The representative protocol uses $\Gamma = 1$ and frequent detection with partial tracing. The first prespecified independently generated pool is shown. Points are ABM means with pointwise normal-approximation 95% confidence intervals, solid curves are the $K = 40$ hierarchy, and the vertical dotted line marks tracing activation at $\tau = 0.5$. The m_1 summaries are conditional on $I(\tau) > 0$.

Table S9 Matched decomposition errors shown in Fig. S4

Protocol	Γ	E_U	E_I	E_{M_1}	E_{mean}
Frequent partial	1	0.000898	0.000837	0.000900	0.001522
Less-frequent complete	1	0.001840	0.001109	0.000833	0.002304

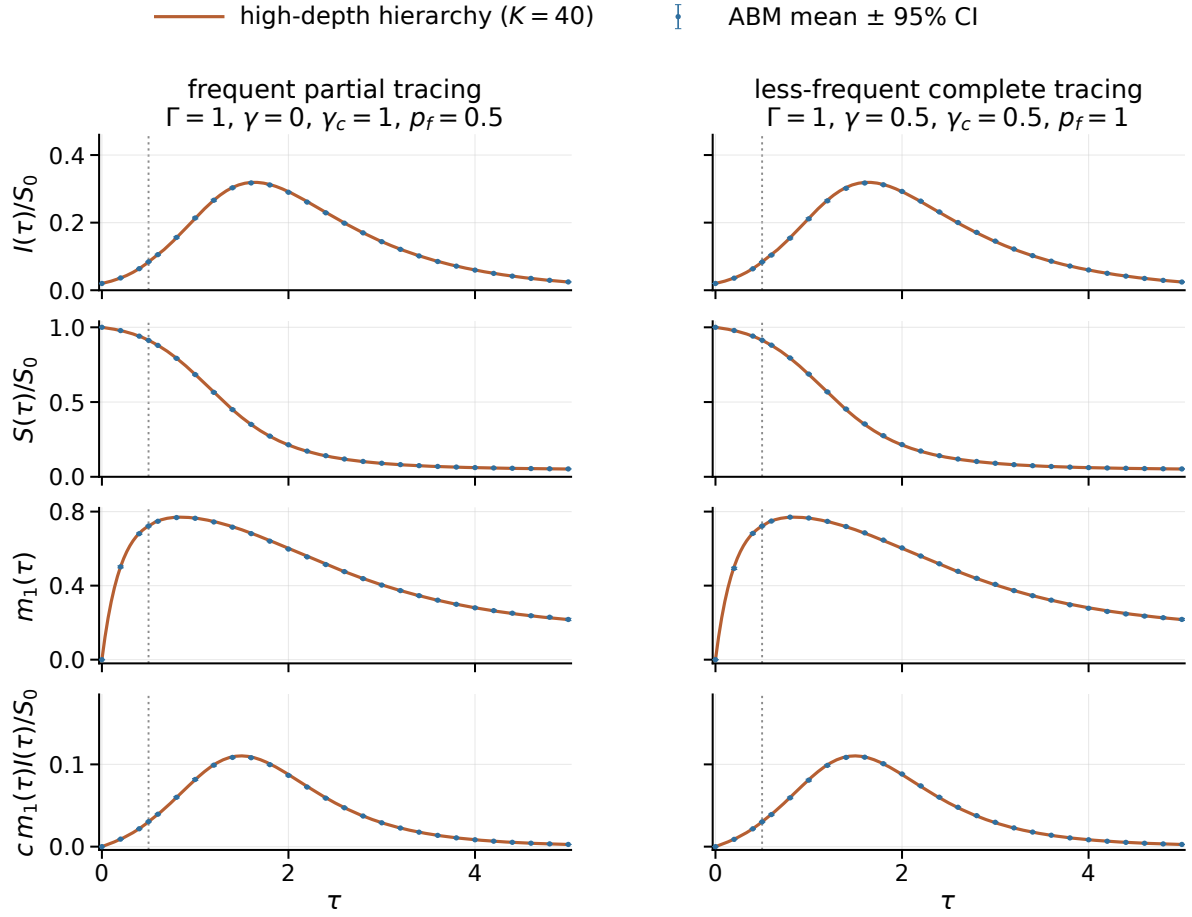


Fig. S4 Alternative detection–tracing decompositions at $R = 4$, $c = 0.5$, and $\Gamma = 1$. Both protocols share the same nondimensional deterministic reference but differ in detection frequency and tracing probability. The first prespecified independently generated pool is shown for each protocol. Points are ABM means with pointwise normal-approximation 95% confidence intervals; the m_1 summaries are conditional on $I(\tau) > 0$.

Table S10 Directly simulated rate-scale protocols shown in Fig. S5

Γ	β	γ	γ_c	p_f	E_{mean}
0.25	0.0001250	0	0.25	0.5	0.000723
0.5	0.0002500	0	0.5	0.5	0.000936
1	0.0005000	0	1	0.5	0.001522
2	0.0010000	0	2	0.5	0.001686
4	0.0020000	0	4	0.5	0.001132

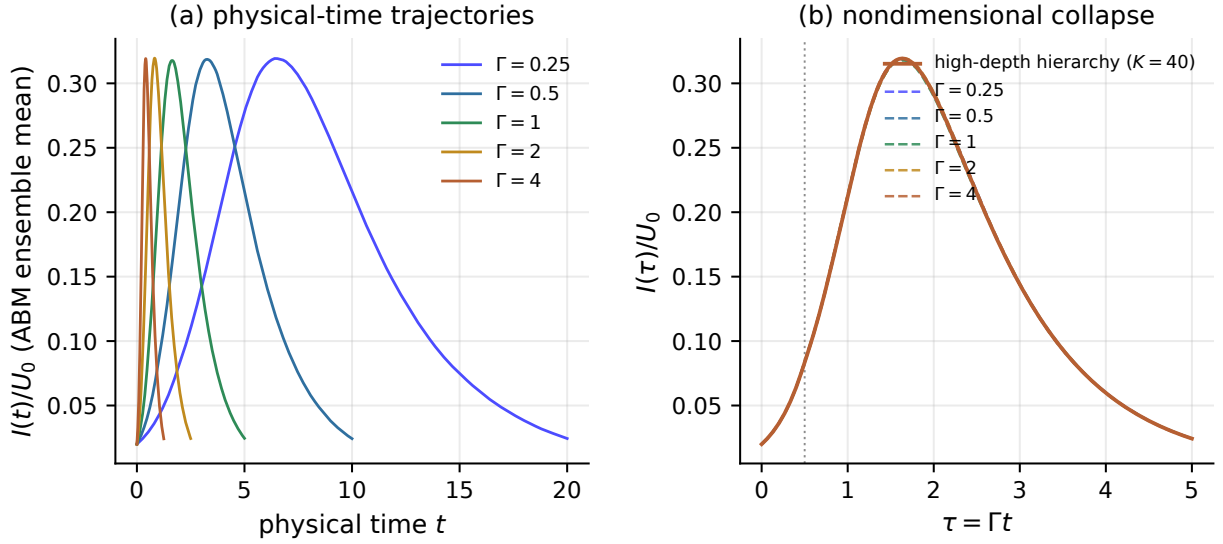


Fig. S5 Direct simulation of dimensional rate rescaling. Independently generated 120-realization ABM ensembles are shown for $\Gamma \in \{1/4, 1/2, 1, 2, 4\}$ at $R = 4$ and $c = 0.5$. The first prespecified pool is used for each rate scale. Panel (a) shows physical-time trajectories over $0 \leq t \leq 5/\Gamma$; proportional changes in the dimensional event rates alter the physical time scale. Panel (b) shows the same ABM ensemble means in $\tau = \Gamma t$, where they collapse closely onto the common high-depth reference. This is an implementation and nondimensionalization check rather than an additional biological sensitivity analysis.

S5.6 Direct event-level verification of the balance terms

The trajectory comparison tests the combined evolution of U , I , and M_1 . To verify the individual balance terms directly, an instrumented version of the same event-driven ABM also records transmission events, spontaneous removals, case-identification events, and the number of infectious direct infectees removed through tracing. Traced removals are counted individually and occur as consequences of identification events; they do not constitute a separate Gillespie clock.

For realization ℓ , the event intensities per unit dimensionless time are

$$q_{\text{trans}}^{(\ell)} = \frac{\beta}{\Gamma} U^{\text{ABM},(\ell)} I^{\text{ABM},(\ell)}, \quad q_{\text{rec}}^{(\ell)} = \frac{\gamma}{\Gamma} I^{\text{ABM},(\ell)},$$

$$q_{\text{id}}^{(\ell)} = \frac{\gamma_c}{\Gamma} I^{\text{ABM},(\ell)}, \quad q_{\text{trace}}^{(\ell)} = \mathbf{1}_{\{\tau \geq \tau_{\text{on}}\}} \frac{\gamma_c p_f}{\Gamma} M_1^{\text{ABM},(\ell)}.$$

For each interval $B_j = [\tau_j, \tau_{j+1})$, the observed ensemble-mean count is $n^{-1} \sum_{\ell} \Delta N_{e,\ell j}$ and the state-supplied analytical expectation is $n^{-1} \sum_{\ell} \int_{B_j} q_e^{(\ell)}(\tau) d\tau$. The state integrals are accumulated exactly over the piecewise-constant ABM holding intervals and are split at the observation boundaries and at $\tau_{\text{on}} = 0.5$. Thus, the analytical axis is formed from the measured ABM states on the same paths and does not use a deterministic reference trajectory or a midpoint approximation.

The analysis uses $U_0 = 8000$, $I_0 = 160$, the full set of 195 raw-rate protocols, and 120 independently generated realizations per protocol. The interval $0 \leq \tau \leq 5$ is divided into 150 equal bins. There are therefore 29,250 protocol-interval comparisons for transmission, spontaneous removal, and identification. The traced-removal panel contains 20,250 positive-hazard comparisons; 9000 zero-hazard intervals were verified separately, and none contained a traced removal.

The aggregate observed-to-expected ratios were 1.00002 for transmission, 1.00009 for spontaneous removal, 0.99997 for identification, and 1.00010 for traced removal.

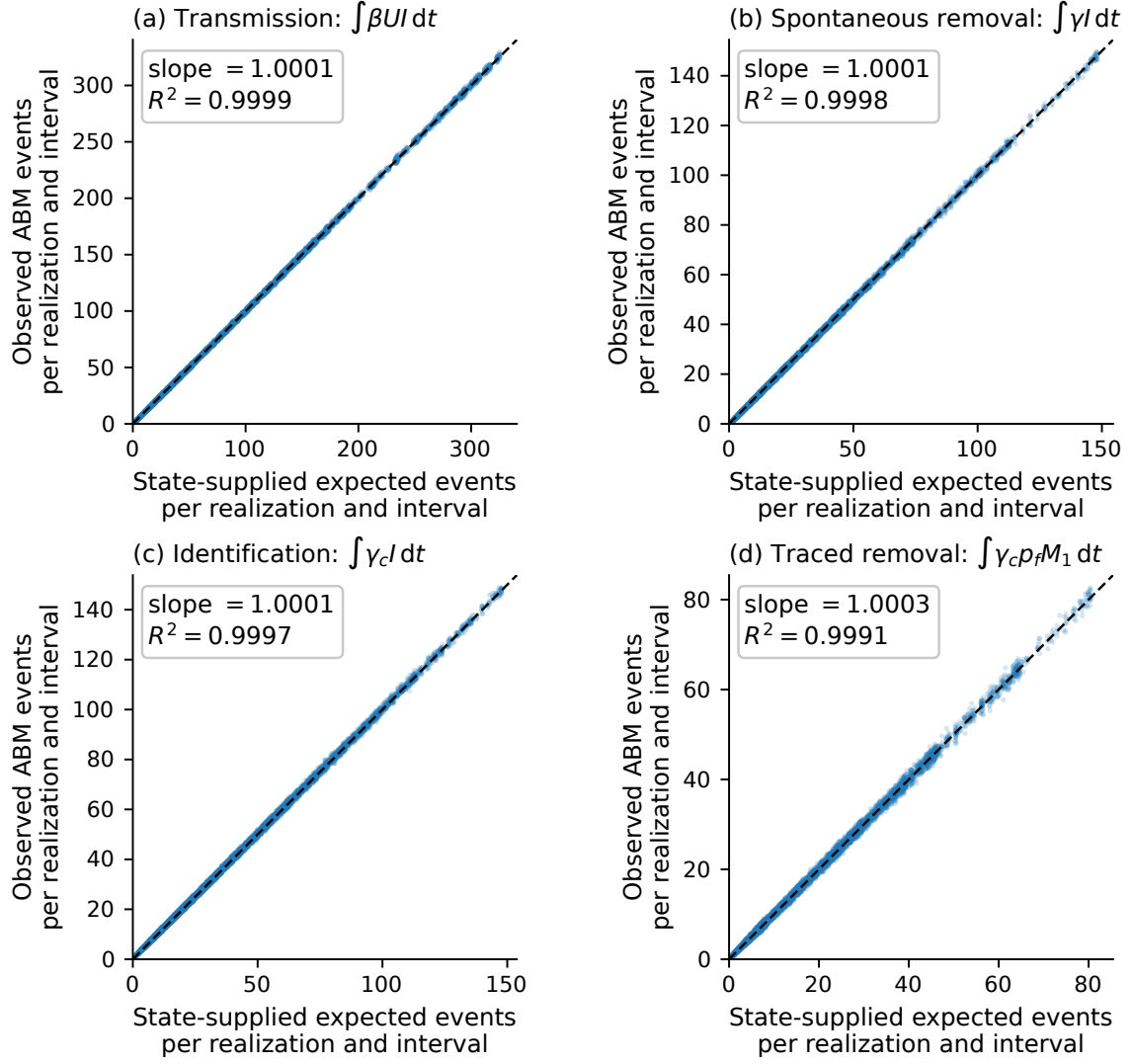


Fig. S6 Direct event-level verification of the four contributions entering the infectious-population balance. The instrumented Gillespie ABM records transmission events, spontaneous removals, identification events, and the number of infectious direct infectees removed by tracing. For the same realized paths, U , I , and M_1 are integrated over each observation interval to evaluate $\int \beta UI dt$, $\int \gamma I dt$, $\int \gamma_c I dt$, and $\int \gamma_c p_f M_1 dt$. Each point represents one of 195 raw-rate protocols and one of 150 intervals, averaged over 120 independent realizations. The dashed line is the identity. Intervals with identically zero tracing intensity were verified separately and omitted from panel (d) to avoid overplotting at the origin. This is a conditional-intensity and implementation check based on ABM-supplied states, rather than an independent fit to the deterministic trajectory.

S6 Closure validation

This section supports the closure validation reported in Section 5.2 of the main text: first the identity that isolates the error a closure introduces, then the figures measuring it against the ensemble ABM.

S6.1 Snapshotwise closure-defect identity

Let $K \geq 1$ and let $z_K = (S, I, m_1, \dots, m_K)$ with $I > 0$ denote the retained state, and suppose the first unresolved coordinate is replaced by an arbitrary closure function, $m_{K+1} \approx \Phi_{K+1}(z_K)$. At the

same retained state and the same tracing coefficient c , the unclosed final-moment equation is

$$m'_{K,\text{uncl}} = R_S(m_{K-1} - m_K) - Km_K + c(m_1m_K - m_{K+1}), \quad (\text{S59})$$

whereas the closed equation replaces m_{K+1} by $\Phi_{K+1}(z_K)$. Subtracting,

$$\Delta_K = m'_{K,\text{uncl}} - m'_{K,\text{cl}} = c[\Phi_{K+1}(z_K) - m_{K+1}], \quad (\text{S60})$$

since the transmission contribution, the primary-removal contribution, and the retained tracing contribution cm_1m_K cancel exactly. Because m_{K+1} occurs only in the equation for m_K , no other retained equation is directly modified.

At observation time τ_j , writing $z_K^{\text{ABM}}(\tau_j)$ for the ensemble-mean retained state and $m_{K+1}^{\text{ABM}}(\tau_j)$ for the corresponding ensemble-mean unresolved coordinate, the snapshotwise ensemble-ABM closure defect is

$$\hat{\Delta}_K^{\text{ABM}}(\tau_j) = c(\tau_j) [\Phi_{K+1}(z_K^{\text{ABM}}(\tau_j)) - m_{K+1}^{\text{ABM}}(\tau_j)], \quad (\text{S61})$$

where $c(\tau_j)$ follows the tracing-activation schedule used in the numerical experiments. This quantifies the local right-hand-side error introduced by the closed term at each observation time. Because the closure map may be nonlinear, it represents the closure evaluated at the ensemble-mean state rather than the ensemble mean of realization-specific defects. For the zeroth-order algebraic QSS closure and the depth- K dynamic closure respectively, the defects are

$$\hat{\Delta}_{\text{QSS}}^{\text{ABM}}(\tau) = c(\tau) [\hat{m}_1^{(0)}(R_S^{\text{ABM}}(\tau)) - m_1^{\text{ABM}}(\tau)], \quad (\text{S62})$$

$$\hat{\Delta}_K^{\text{ABM}}(\tau) = c(\tau) [\Phi_{K+1}^{\text{tail}}(R_S^{\text{ABM}}(\tau); m_K^{\text{ABM}}(\tau)) - m_{K+1}^{\text{ABM}}(\tau)], \quad K = 1, 2, 3. \quad (\text{S63})$$

The corresponding scatter and distributions are shown in Fig. S7.

S6.2 Closure validation figures

Fig. S7 compares the closure-supplied local term with the corresponding term measured from the ensemble ABM at the same observation time, and Fig. S8 shows the depth- K dynamic-closure trajectory overlays at $R = 4$ that accompany the algebraic-closure overlays displayed in the main text.

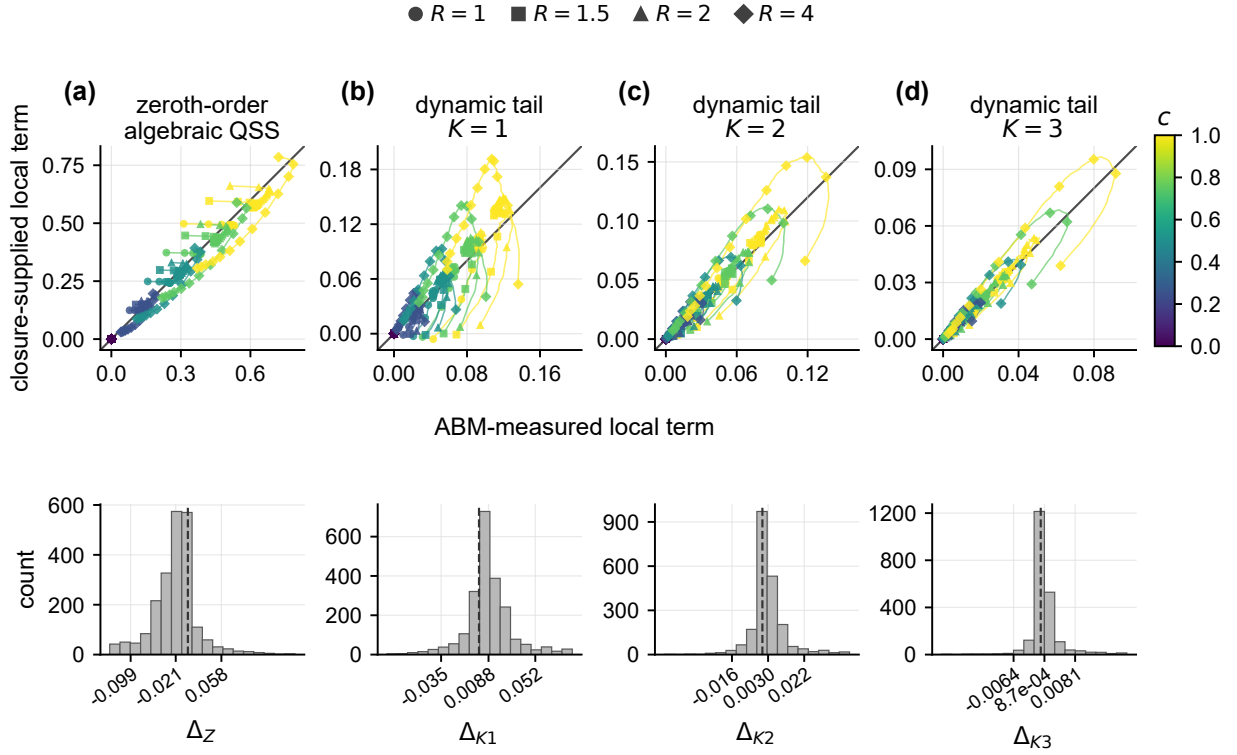


Fig. S7 Local closure terms evaluated along ensemble-mean finite-population ABM trajectories. Each colored path comprises post-activation snapshots ($\tau \geq 0.5$) obtained after averaging 120 independent realizations within each parameter cell; color denotes c and marker shape denotes R . In the top row, the horizontal coordinate is the ABM-measured local term and the vertical coordinate is the closure-supplied local term; the diagonal denotes equality. (a) Zeroth-order algebraic QSS closure: $c\hat{m}_1^{(0)}(\bar{R}_S^{\text{ABM}})$ is compared with $c\bar{m}_1^{\text{ABM}}(\bar{R}_S^{\text{ABM}})$. (b–d) Dynamic tail closures $K = 1, 2, 3$: $c[\bar{m}_1^{\text{ABM}}\bar{m}_K^{\text{ABM}} - \bar{m}_{K+1}^{\text{ABM}}]$ is compared with $c[\bar{m}_1^{\text{ABM}}\bar{m}_K^{\text{ABM}} - \Phi_{K+1}^{\text{tail}}(\bar{R}_S^{\text{ABM}}; \bar{m}_K^{\text{ABM}})]$. All quantities in a snapshot are evaluated at the same ensemble-mean state, with $\bar{R}_S^{\text{ABM}} = R\bar{S}^{\text{ABM}}/S_0$. The bottom row shows descriptive snapshot distributions of the signed closure-supplied-minus-ABM-measured differences, labeled Δ_Z , Δ_{K1} , Δ_{K2} , and Δ_{K3} . Histograms pool the displayed ensemble-mean snapshots with $c > 0$; their three x-axis ticks are the mean minus two population standard deviations, the mean, and the mean plus two population standard deviations, rounded to two significant digits. Simulations use $S_0 = 8000$, $I_0 = 160$, $R \in \{1, 1.5, 2, 4\}$, $c \in \{0, 0.25, 0.5, 0.75, 1\}$, and 151 observation times on $0 \leq \tau \leq 5$, with $(\gamma/\Gamma, \gamma_c/\Gamma, p_f) = (0, 1, c)$. Normalized genealogical coordinates are averaged conditionally over realizations with positive infectious count.

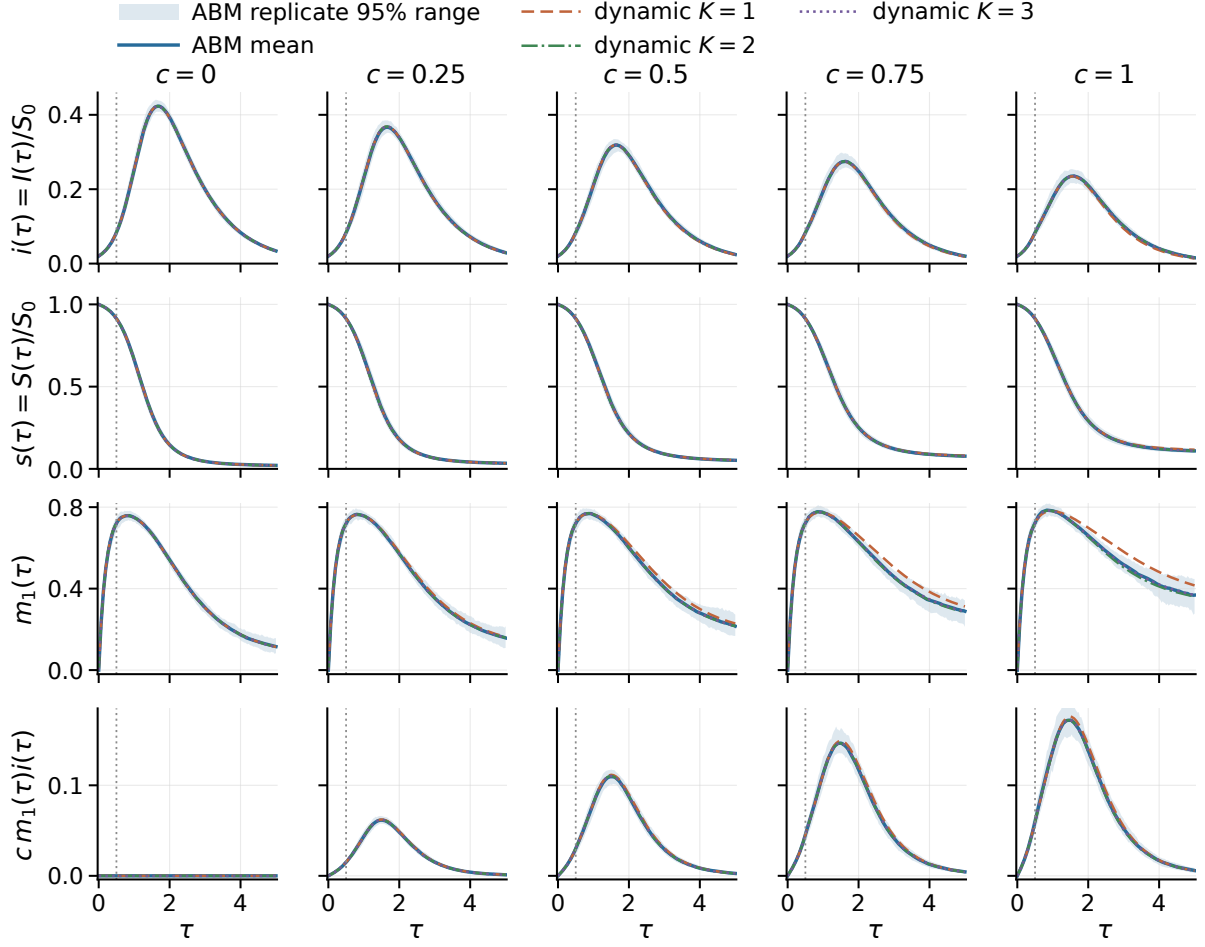


Fig. S8 Representative finite-pool trajectories for the depth- K dynamic closures at $R = 4$. Columns show $c = 0, 0.25, 0.5, 0.75$, and 1 ; rows show $i = I/S_0$, $s = S/S_0$, $m_1 = M_1/I$, and the normalized deterministic tracing flux $c m_1(\tau) i(\tau)$. Blue curves are the sample ensemble-mean ABM trajectory and pale bands are pointwise 2.5th–97.5th replicate percentiles across 120 independent realizations. Orange dashed, green dash-dotted, and purple dotted curves are the dynamic $K = 1, 2$, and 3 closures. The vertical dotted line marks activation of forward tracing at $\tau = 0.5$. Simulations use $S_0 = 8000$, $I_0 = 160$, and 151 observation times on $0 \leq \tau \leq 5$, with $(\gamma/\Gamma, \gamma_c/\Gamma, p_f) = (0, 1, c)$. These are the same $R = 4$ ABM replicates underlying the main-text closure-trajectory errors and algebraic-closure overlays.

S7 Reproducibility files

Supplementary Material 2 provides the dimensional parameter table, the published error and convergence summaries, and the deterministic-reference and event-term diagnostics. Supplementary Material 3 provides the executable and audit code, the numerical and figure data, the figure-generation files, and the software requirements.

Within Supplementary Material 3, the material behind Sections S2 and S3 comprises the complete Fig. S1 curve data, the maximum-error summary, the symbolic coefficient audit, the QSS solver diagnostics, and standalone generation scripts, stored under `data/`, `diagnostics/`, and `reproducibility/`; the symbolic audit independently verifies the displayed coefficients.

References

The continued-fraction and special-function numerical procedures in Section S2.2 use Gautschi (1967) and Olver et al. (2010).

Amos DE (1974) Computation of modified Bessel functions and their ratios. *Math Comput* 28:239–251. <https://doi.org/10.1090/S0025-5718-1974-0333287-7>

Gautschi W (1967) Computational aspects of three-term recurrence relations. *SIAM Rev* 9:24–82. <https://doi.org/10.1137/1009002>

Krantz SG, Parks HR (2002) The implicit function theorem: history, theory, and applications. Birkhäuser, Boston. <https://doi.org/10.1007/978-1-4612-0059-8>

Murdock JA (1999) Perturbations: theory and methods. *Classics in Applied Mathematics*, vol 27. SIAM, Philadelphia. <https://doi.org/10.1137/1.9781611971095>

Olver FWJ, Lozier DW, Boisvert RF, Clark CW (eds) (2010) NIST handbook of mathematical functions. Cambridge University Press, New York. <https://www.nist.gov/publications/nist-handbook-mathematical-functions>